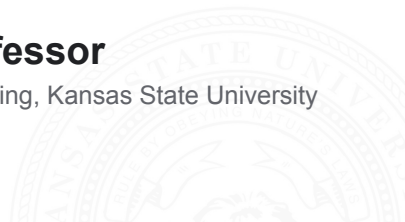


Course Orientation & Logistics

Buxin She | Assistant Professor

Mike Wieggers Department of Electrical and Computer Engineering, Kansas State University



Meet your instructor

EDUCATION AND PROFESSIONAL EXPERIENCE



2020–2023

Ph.D., Electrical Engineering
University of Tennessee, Knoxville

2022

Research Intern
Argonne National Laboratory

2023–2025

(Senior) Research Engineer
Pacific Northwest National Laboratory

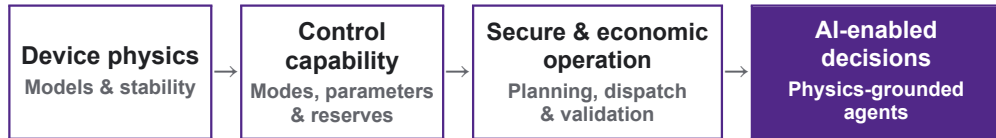
2026–now

Assistant Professor
Kansas State University

Research Mission

INTELLIGENT POWER (IP) LAB AT KSU

Advance power systems through inverter-based resources (IBRs) and artificial intelligence (AI) to enable secure, resilient, and economical power supply.



Meet your teaching assistant



Ryan Hoyt

Teaching Assistant
rchoyt@ksu.edu

Background

- 1 K-State B.E.E. candidate:** Electrical Engineering with a Leadership Studies minor; expected December 2026.
- 2 Evergy, Long-Term Transmission Planning:** line-impedance calculations and NERC compliance support.
- 3 Summer 2026, Generation Development:** supporting new power-plant projects through field visits and technical work.

Course overview

COURSE OVERVIEW

Scope

This course introduces the fundamental concepts of electric power systems engineering.

Primary focus

Modeling power-system components and studying an interconnected system under steady-state conditions.

The course moves from individual components to a defensible system-level recommendation.

- 1 Describe the physical equipment and network.
- 2 Build consistent mathematical models.
- 3 Use analysis to support a design decision.

Course goals

COURSE GOALS AND LEARNING OUTCOMES

- 1 Compute power flows in three-phase circuits.
- 2 Model transformers and transmission lines.
- 3 Solve power systems using the per-unit method.
- 4 Compare power-generation alternatives economically.
- 5 Perform and interpret PowerWorld power-flow simulations.
- 6 Recognize social and environmental issues associated with power systems.

Reading

Required text

Power System Analysis and Design

J. D. Glover, T. J. Overbye, M. S. Sarma,
and A. B. Birchfield
Seventh Edition, Cengage, 2023

Chapters 1–6 plus the Optimal Fuel Mix
supplement.

Additional references

- J. J. Grainger and W. D. Stevenson Jr., *Power System Analysis*, McGraw-Hill, 1994.
- C. A. Gross, *Power System Analysis*, 2nd ed., Wiley, 1986.
- G. W. Stagg and A. H. El-Abiad, *Computer Methods in Power System Analysis*, McGraw-Hill, 1968.

Course outline

Foundations

Introduction - Chapter 1

Optimal Fuel Mix - Supplement

Single- and Three-Phase AC - Chapter 2

Equipment models

Transformer Modeling - Chapter 3

Per-Unit System - Chapter 3

Three-Phase Transformers - Chapter 3

Network models

Transmission-Line Parameters - Chapter 4

Transmission-Line Models - Chapter 5

System solution

Power Flow - Chapter 6

PowerWorld studies - guided course cases

Suggested problems

Chapter	Suggested problems
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2	1–7, 10–11, 19–21, 24, 27, 30, 40–43, 45, 48, 51
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3	20–23, 26–28, 41, 49
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4	None
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5	1, 2, 26, 28, 32, 48–50
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6	7, 24, 28, 30, 32, 34, 36–38
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These are practice targets, not an assignment list unless a submission is posted in Canvas.

Course logistics

WHERE AND HOW WE MEET

In person

**Monday / Wednesday
/ Friday**

12:30–1:20 PM

DUE 0097

Expected effort

Plan for 8–10 hours each week beyond lecture.

Canvas

Announcements, materials, assignments, submissions, grades, and course instructions.

October 12 and 14

No onsite class. Follow the video or other instructions posted in Canvas.

Getting help

COMMUNICATION AND SUPPORT

Course questions

Canvas discussion

Use it when the answer may help other students. Keep one topic per post and use a clear subject line.

Private questions

Email

Instructor: bshe@ksu.edu

TA: rchoyt@ksu.edu

Subject: Name | ECE 685 | Topic

Office hours

Tuesday and Thursday, 10:00 AM–12:00 PM. Email to request an appointment; other times are possible when available.

Response target

Within 24 hours on weekdays and 48 hours on weekends for email or course communication.

Course tools

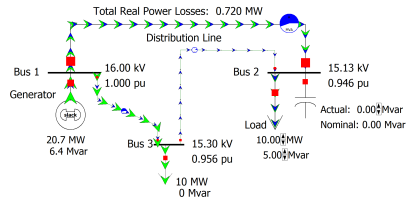
COMPUTER AND TECHNOLOGY USE

Canvas

- announcements and instructions;
- slides, assignments, and submissions;
- grades and class communication.

Technology

Keep your browser and PDF reader current. Report technical failures to the K-State IT Help Desk.



PowerWorld will build, solve, and test steady-state network models.

Grading

ASSESSMENT WEIGHTS

Assessment	Points
Exam 1	100
Exam 2	100
Comprehensive Final	100
Design Project	100
Homework	50
Attendance	20
Total	470

Baseline cutoffs

A : 90 *B* : 80

C : 70 *D* : 60

- cutoffs may be adjusted based on exam nature and performance;
- appropriate project performance is considered separately.

Workload and practice

EXAMS, HOMEWORK, ATTENDANCE, AND WORKLOAD

- 1 **Exams:** two in-class semester exams plus a comprehensive finals-week exam.
- 2 **Project:** progress evidence and a formal final report.
- 3 **Homework:** practice and preparation submitted as directed.
- 4 **Attendance:** regular participation is part of the course grade.

Give yourself time to practice consistently, and ask questions early before confusion compounds.

Early-semester dates

AUGUST THROUGH OCTOBER

- 1 **Aug. 24:** classes begin.
- 2 **Sep. 7:** Labor Day - no class.
- 3 **Oct. 2:** Exam 1.
- 4 **Oct. 9:** Wildcat Pause Day.
- 5 **Oct. 12 and 14:** no onsite class; follow Canvas instructions.

Final-month dates

NOVEMBER AND DECEMBER

- 1 **Nov. 13:** Exam 2.
- 2 **Nov. 22–29:** Fall Break.
- 3 **Dec. 9:** Final Project due.
- 4 **Dec. 11:** last day of classes.
- 5 **Dec. 16:** comprehensive Final.

The official K-State calendar controls campus-wide dates; Canvas will carry course-specific corrections.



Problems and late work

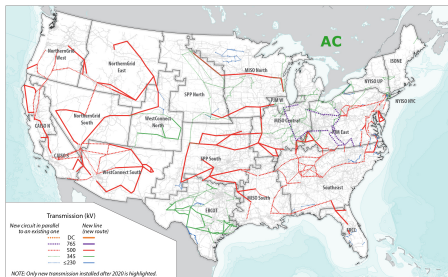
SUBMISSION, LATE WORK, AND TECHNICAL DIFFICULTIES

- 1 Submit through the Canvas Dropbox or as a physical copy when directed.
- 2 Most deadlines are in class or 11:59 PM Central on the stated date.
- 3 If you cannot finish on time, email as soon as possible - not after silence.
- 4 Document technology failures with the K-State IT Help Desk.

Exceptions are considered case by case; timely communication and documentation matter.

Final Project

GENERATION-AND-TRANSMISSION DESIGN



Design question

Recommend a technically feasible and economical generation-and-transmission plan.

- choose generation;
- choose network reinforcements;
- justify reliability and cost.

Responsible AI use

CONSTRUCTIVE USE AND REQUIRED CHECKS

Encouraged

- clarify concepts;
- explore alternative approaches;
- support coding, writing, and checking.

Required

- verify every result;
- explain every submitted claim;
- disclose the tool and its contribution.

AI may accelerate a step; it may not replace your understanding.

Prohibited AI use

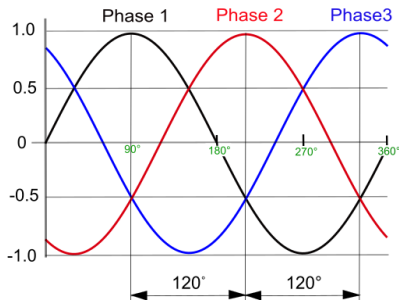
ACADEMIC INTEGRITY

- 1 Treating an AI answer as automatically correct.
- 2 Submitting material AI contribution without disclosure.
- 3 Using unauthorized AI assistance on an exam or restricted assignment.

A useful disclosure names the tool, the task it supported, and how the result was verified.

Three-phase foundations

PREREQUISITE FOUNDATIONS

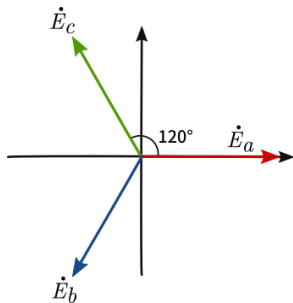


EECE 581 foundation

- AC steady-state analysis;
- phasors and complex power;
- balanced three-phase circuits;
- transformer and synchronous-machine models;
- matrix fundamentals.

Prerequisite check

PREREQUISITE READINESS



Balanced three-phase
voltage phasors

- 1 Can I interpret magnitude and phase angle?
- 2 Can I distinguish MW, Mvar, and MVA?
- 3 Can I analyze a balanced three-phase circuit?
- 4 Can I read a transformer equivalent circuit?
- 5 Can I set up a small matrix equation?

Power-system history

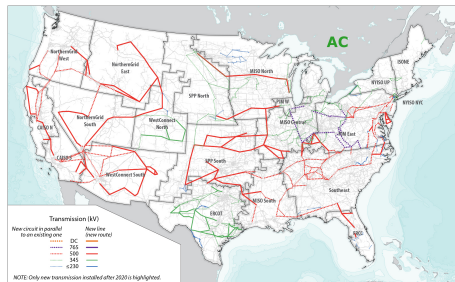
A SHORT HISTORICAL PERSPECTIVE

1882 - Pearl Street

A local DC system served 85 customers within roughly one square mile.

1884 - Early AC transmission

Transformer-based AC demonstrated practical power transfer over a much longer distance.



Today's interconnected networks coordinate many generators, transmission paths, and loads across regions.

Physical power system

THE PHYSICAL PATH



measurement, communication, protection, and control

The grid is an energy-delivery system and an information-and-control system at the same time.

Grid transformation

TRADITIONAL STRUCTURE AND EMERGING OPERATION

Traditional structure

- large central generating plants;
- predominantly one-way delivery;
- limited visibility near end users.

Evolving structure

- renewable generation and storage;
- bidirectional power flows;
- more sensors, communication, and automated control.

New technology changes where power enters the network and how operators observe and control it.

Reliability challenge

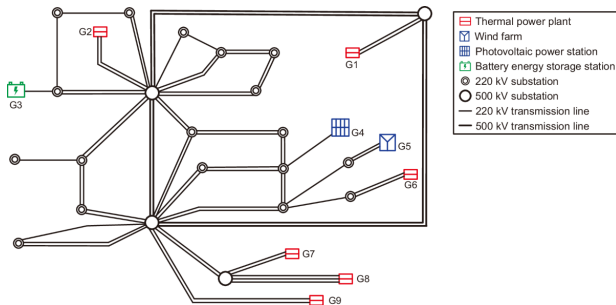
THE ENGINEERING CHALLENGE

- 1 Balance electricity production and consumption continuously.
- 2 Keep voltages, currents, and equipment loading within limits.
- 3 Withstand equipment outages, weather, and other disturbances.
- 4 Adapt generation and transmission economically as needs change.

Every grid transition must preserve reliable service while expanding capability.

Design workflow

BRIDGE TO LECTURE 02



- 1 Represent** the connected physical system with consistent data.
- 2 Solve** the mathematical model under defined conditions.
- 3 Check** limits, balances, and operating credibility.

Lecture 02 begins with the equipment and one-line vocabulary used to build that model.